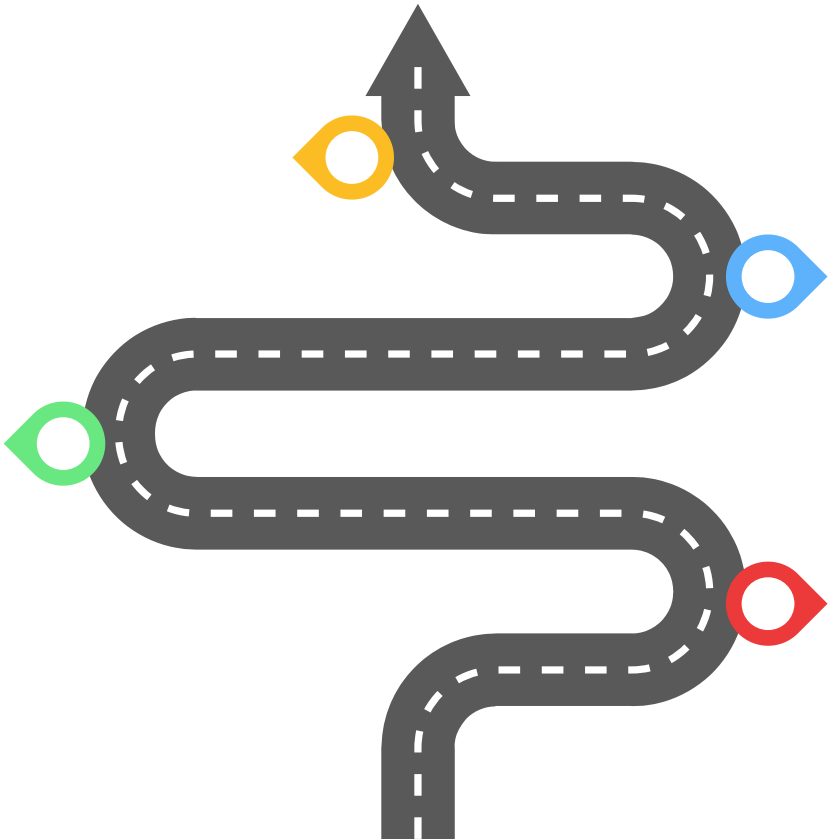
Uber Wait Times

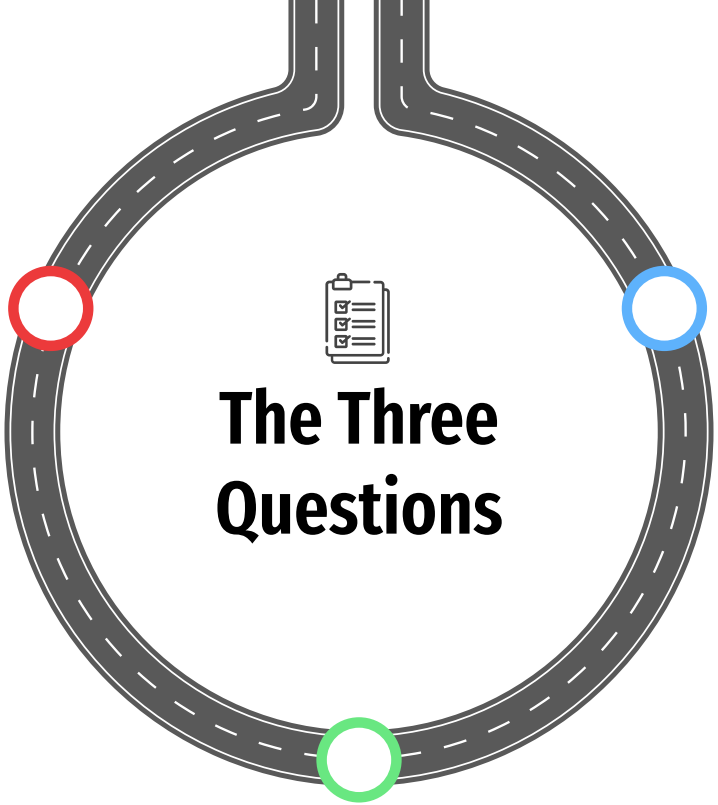
in the Philadelphia Area



Problem Kernel

Uber wait times can be very unpredictable, and wait times can be longer than expected, especially during large events such as sports games and concerts

The Three Questions



1.

How should I place a value on the different outcomes my model will produce?

2.

What are the critical decisions I should include in the model?

3.

Which factors are most uncertain, and which will have the biggest impact on the outcome?

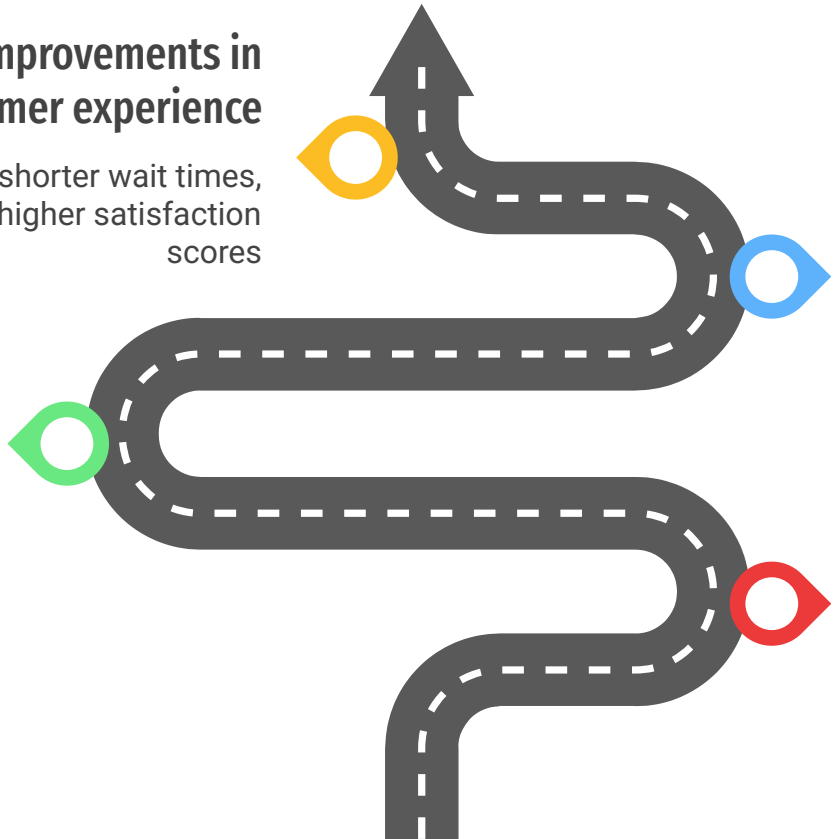
Value of the Different Outcomes the Model will Produce

Quantify improvements in customer experience

Measure value through shorter wait times, fewer cancellations, and higher satisfaction scores

Measure operational efficiency gains

Value outcomes by reductions in wasted time, smoother traffic flow, and more effective matching between riders and drivers



Quantify improvements in driver experience

Value outcomes by increases in driver earnings, reduced wait time, and more predictable work patterns

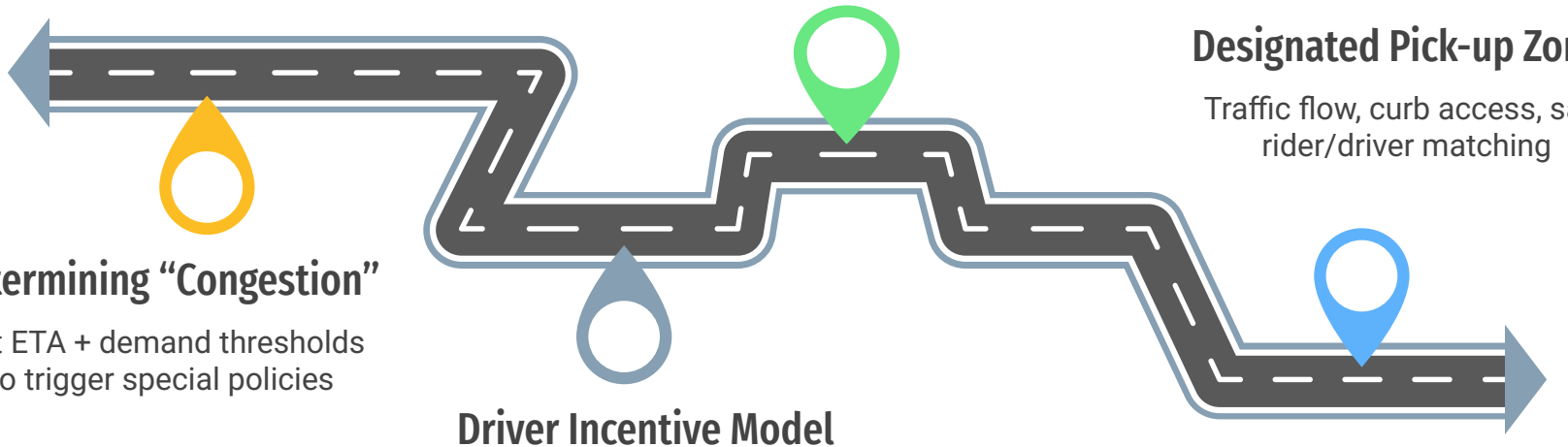
Estimate financial impact

Value outcomes through increases in completed trips, higher revenue, and better long-term user retention

Critical Decisions included in the Model

Pre-Positioning Drivers

Staging around Citizens Bank Park, the Linc, Xfinity, Stateside Live



Designated Pick-up Zones

Traffic flow, curb access, safe rider/driver matching

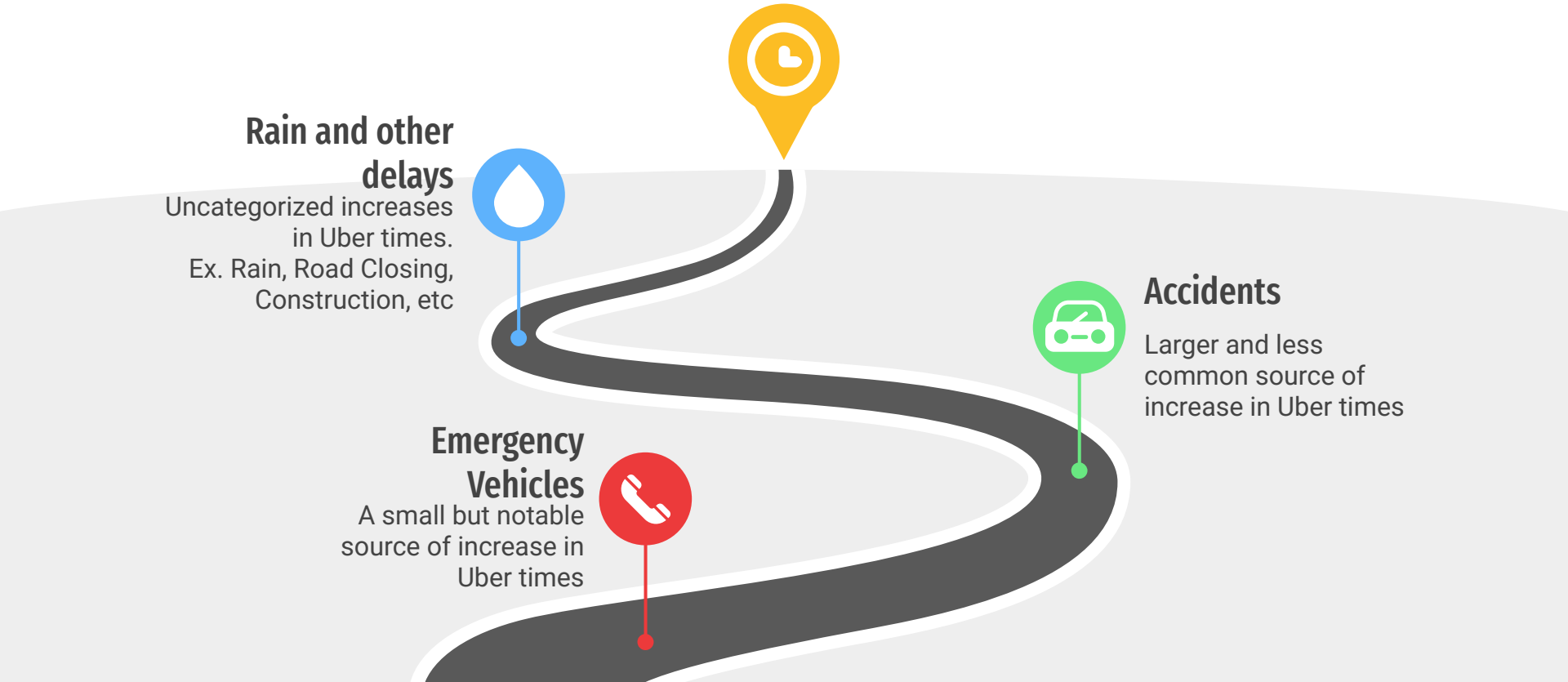
Determining “Congestion”

Set ETA + demand thresholds to trigger special policies

Driver Incentive Model

Bonuses, guaranteed earnings, or capped surge pricing

Most Uncertain Factors with Biggest Impact Outcome



Rain and other delays

Uncategorized increases in Uber times. Ex. Rain, Road Closing, Construction, etc

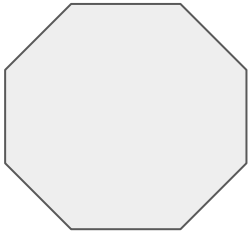
Accidents

Larger and less common source of increase in Uber times

Emergency Vehicles

A small but notable source of increase in Uber times

Model Components



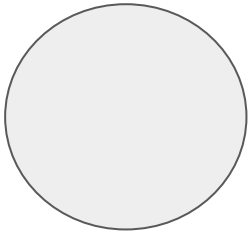
Outcome Measure

- Total Time for an Uber per person



Parameters

- Distance to Pickup
- Distance to Destination



Intermediate Variables

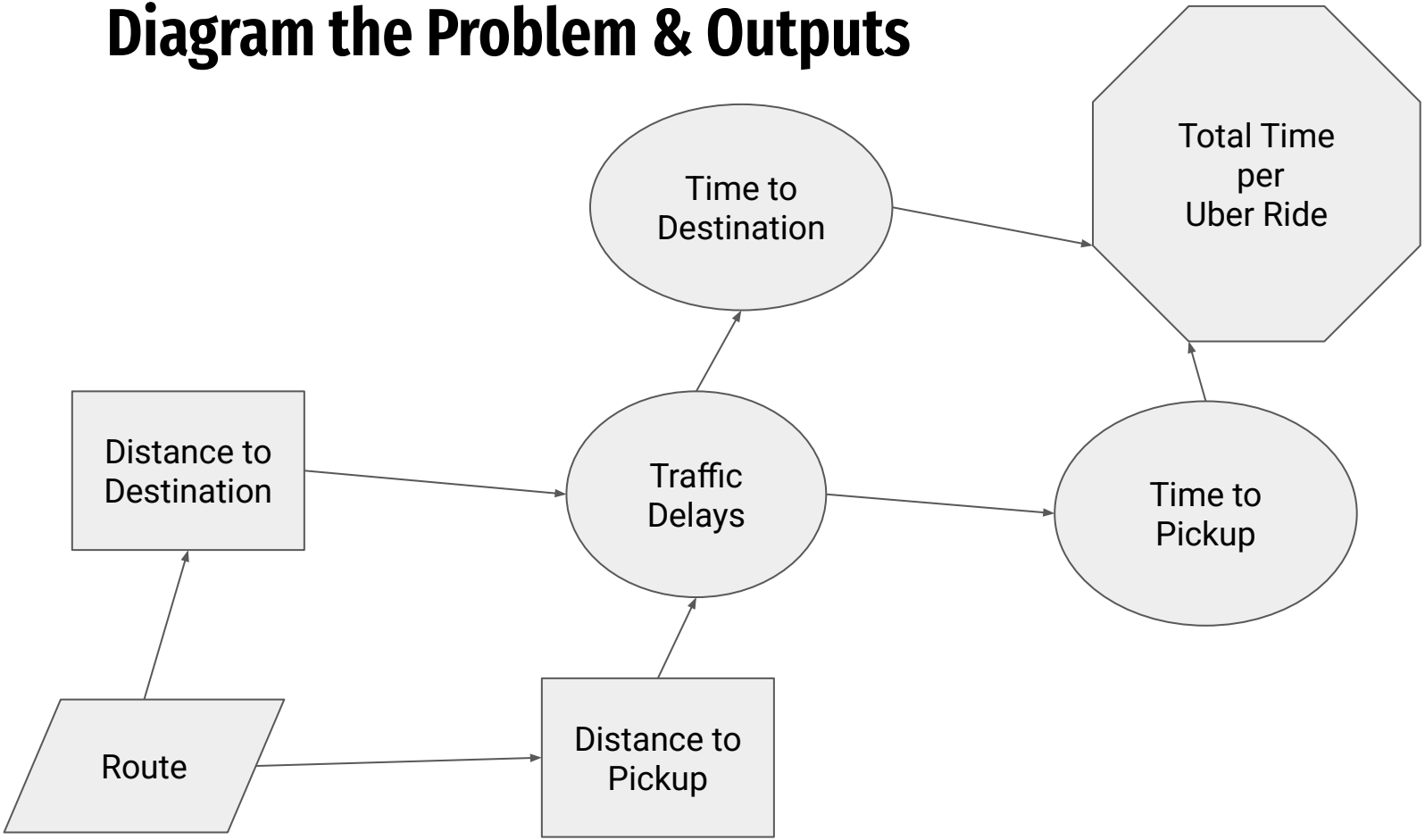
- Time to Pickup
- Time to Destination
- All Types Delays



Decision Variables

- Route
- # of passenger

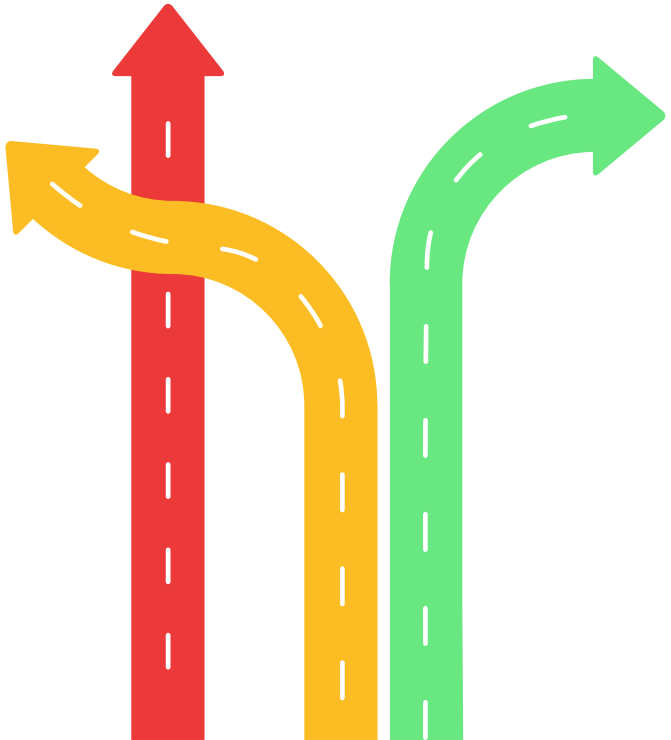
Diagram the Problem & Outputs



Possible Solutions

Create a cost incentive for heavy traffic days

Uber can notify drivers on heavy traffic days



Uber buses/passenger vans

Thank you!

